AI Fracture Detection Prototype

AI prototype

Research and demonstration prototype for AI-based fracture detection using medical imaging data.
Intended for educational and research purposes only.
Not for clinical or diagnostic use.

Collaboration

Open to collaboration with researchers, clinics and organizations involving custom medical imaging datasets, AI model training, medical imaging research, and research prototype development.

Results

The AI software demonstrates strong fracture detection performance across multiple public medical imaging datasets.

On the Mendeley Human Bone Fracture Dataset dataset (1,539 images), the model achieved:

- **Accuracy: 0.97**
- **AUC: 0.9891**
- **F1-score: 0.9836**
- **Brier: ~ 0.023**

These results indicate excellent classification capability, strong predictive confidence, and reliable probability calibration.

On the FracAtlas dataset (4,083 images), the model achieved:

- **Accuracy: 0.9054**
- **AUC: 0.9034**
- **F1: 0.7041**
- **Brier: 0.0768**

The results demonstrate strong generalization across different fracture imaging datasets and consistent performance in fracture classification tasks.

All evaluations were conducted for research and demonstration purposes only and are not intended for clinical or diagnostic use.